LOANZO PROTOCOL: FORMAL UML USE CASE SPECIFICATION

System Architecture, OMG UML 2.5 Models, GeeksforGeeks Formal Specifications & QA Verification Matrix

Standard Compliance: IEEE Std 830-1998, OMG UML 2.5	**Software Domain:** FinTech / P2P Micro-lending
Design Methodology: Use-Case-Driven Development (UCDD)	**QA Framework:** Equivalence Partitioning & BVA
Statutory Reference: NI Act 1881 (Sec 4), Order 37 CPC	**Platform Version:** v2.4 Release Candidate

1. Executive Summary & OMG UML Actor Hierarchy

This specification establishes the formal software engineering UML Use Case modeling foundation for the Loanzo Peer-to-Peer Micro-lending Platform. In alignment with **GeeksforGeeks (GFG)** software development standards, Pressman Software Engineering principles, and IEEE Std 830-1998, all functional capabilities are partitioned across well-defined system boundaries, strict actor specializations, and OMG UML relationship semantics.

Actor Name	Type & Multiplicity	Role Description	Interactions
PlatformMember	Abstract Primary	Base entity for authenticated users with registered cryptographic keys.	Authenticate, Manage Profile, Push Alerts
Borrower	Primary (1..*)	Specializes PlatformMember. Applies for micro-credit, pledges gold/vehicle collateral.	Request Loan, Pledge Collateral, e-Sign Note, Pay EMI
Lender	Primary (1..*)	Specializes PlatformMember. Underwrites bids, provides capital, manages yields.	Browse Marketplace, Submit Loan Bid, Disburse Funds
FieldAgent	Primary / Custodian	Verifies physical collateral, issues geo-tagged vault deposit receipt tokens.	Appraise Bullion, Store Collateral, Register Vault Token
PlatformAdmin	Primary Admin	Oversees platform liquidity, audit compliance, KYC sanctions, dispute resolution.	Audit Ledgers, Freeze Delinquent Accounts, Sanction Check
NPCI UPI Switch	External Secondary	Unified Payments Interface rail for instant P2P disbursements & autopay e-mandates.	Disburse Capital, Settle Repayments, Handle Chargebacks
DigiLocker / NeSL	External Regulatory	Government-approved identity repository & National E-Governance Services Information Utility.	Aadhaar XML, PAN OCR, Legal Debt Registration

2. Master UML Use Case Architecture Diagram

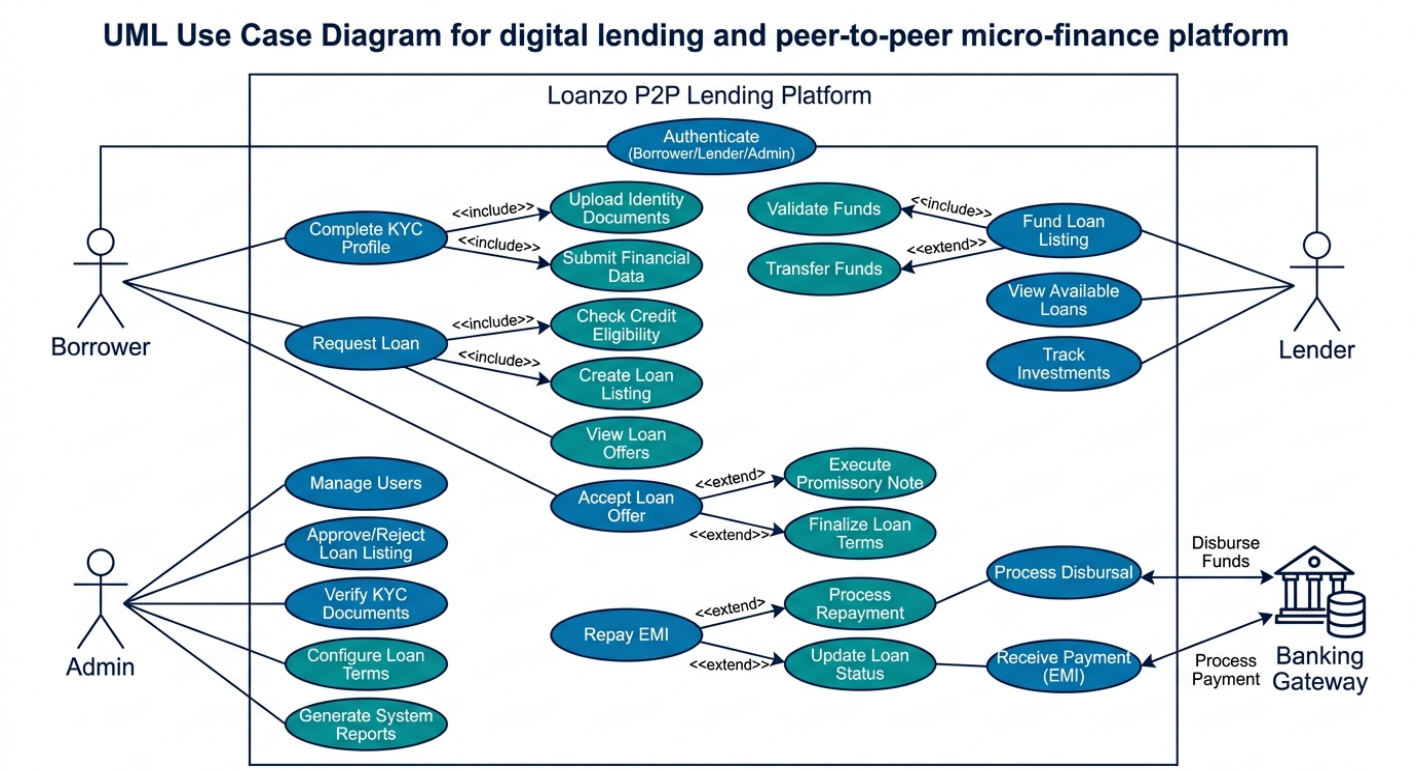


Figure 1: OMG UML 2.5 Use Case Diagram for the Loanzo Digital Micro-Lending Platform (Boundary, Actors, Includes, and Extends)

3. Rigorous UML Relationship Semantics (OMG UML 2.5 vs GFG Practice)

In professional software engineering, misuse of <<include>> and <<extend>> is the #1 design flaw. Loanzo enforces strict OMG UML 2.5 semantic rules:

Relationship Type	Arrow Direction	Execution Condition	Concrete Loanzo Implementation
<<include>>	Base Use Case --> Included UC	**Mandatory**: Base use case cannot complete without executing included UC.	UC-05 (Create Listing) unconditionally includes UC-02 (Verify KYC) and UC-03 (Calculate CIBIL).
<<extend>>	Extension UC --> Base Use Case	**Conditional**: Executes only when specific Extension Point condition is met.	UC-15 (Levy Penalty) extends UC-14 (Pay EMI) at Extension Point: *Past Due Date > 3 Business Days*.
Actor Generalization	Child Actor --|> Parent Actor	**Inheritance**: Child actor inherits all association lines of parent actor.	Borrower and Lender inherit base authentication and session capabilities from PlatformMember.

4. Formal Use Case Specifications (GeeksforGeeks & Cockburn Style)

Representative mission-critical use cases detailed with comprehensive exception and alternative handling.

Use Case ID: UC-05	**Use Case Name: Create Peer-to-Peer Micro-Loan Listing**
Primary Actor: Borrower	**Priority / Risk:** High / Mission Critical
Brief Description: Borrower compiles loan requirements (principal, tenor, interest rate) and publishes a listing to the marketplace.	
Pre-Conditions: Borrower is authenticated via Android StrongBox biometric session; Tier-1 KYC status is ACTIVE; CIBIL >= 500.	
Trigger: Borrower taps 'Create Loan Request' on the Marketplace dashboard.	
Main Success Scenario: 1. System loads Borrower KYC profile and computes borrowing eligibility ceiling. 2. Borrower inputs desired loan amount (INR 1,000 - 500,000) and tenor (1 - 36 months). 3. Borrower selects collateral type (Unsecured, 24K Bullion Gold, or Vehicle Hypothecation). 4. System invokes <<include>> UC-07 (Calculate Collateral LTV) if collateral is pledged. 5. Borrower submits listing; System generates cryptographic listing SHA-256 fingerprint. 6. Listing transitions to STATE: LISTED_PUBLIC.	
Alternative / Exception Flows: • **4a. LTV Exceeded:** If pledged collateral value is < 133% of principal (LTV > 75%), system blocks creation and prompts for additional collateral. • **2a. Max Active Loans Exceeded:** If borrower has >= 3 open loans, system aborts with error code ERR_BORROW_CAP_EXCEEDED.	
Post-Conditions: Listing is persisted in Room SQLite DB, broadcast to active lenders via WebSocket, and timer starts.	

Use Case ID: UC-11	**Use Case Name: Execute Statutory Promissory Note & Agreement**
Primary Actor: Borrower & Lender	**Priority / Risk:** High / Mission Critical
Brief Description: Generates legally enforceable Section 4 Negotiable Instruments Act 1881 promissory note with digital stamp & signature.	
Pre-Conditions: Listing is 100% funded by Lender(s); State is ESCROW_RESERVED.	
Trigger: Borrower taps 'Review & e-Sign Loan Contract'.	
Main Success Scenario: 1. System invokes LoanzoPdfEngine to render dual-party Section 4 NI Act legal contract with precise amortization schedule. 2. System invokes <<include>> UC-12 (Affix Digital E-Stamp Token) via NeSL / State Treasury API. 3. Borrower reviews terms and provides StrongBox Biometric Signature (ECDSA P-256 with user-presence gate). 4. Lender provides matching cryptographic countersignature. 5. System generates immutable SHA-256 digest of final PDF and timestamps it. 6. Loan state transitions to STATE: EXECUTED_DISBURSAL_READY.	
Alternative / Exception Flows: • **3a. Biometric Mismatch / Lockout:** StrongBox returns BiometricPrompt.BIOMETRIC_ERROR_LOCKOUT; user must authenticate via hardware device PIN. • **2a. E-Stamping Gateway Timeout:** System retries via fallback state treasury buffer; queues transaction for offline signing.	
Post-Conditions: Tamper-proof PDF stored in app scoped storage; contract SHA-256 hash registered in NeSL Information Utility.	

5. Software Testing Derivation & Verification Framework

In accordance with software engineering testing methodologies (Pressman Ch. 19, GeeksforGeeks BVA Guide), Use Cases are directly translated into Black-Box Test Cases using **Equivalence Partitioning (EP)** and **Boundary Value Analysis (BVA)**.

Parameter Under Test	Valid Range	Boundaries Evaluated	Expected System Behavior	Kotlin Test Case
Loan Principal (INR)	1,000 - 500,000	Min-1: 999 Min: 1,000 Max: 500,000 Max+1: 500,001	Reject (<1000) Accept Accept Reject (>500k)	`MarketplaceViewModelTest.kt testLoanAmountBoundaries()`

Annual Interest Rate (%)	6.00% - 36.00%	Min-1: 5.99% Min: 6.00% Max: 36.00% Max+1: 36.01%	Reject (Sub-market) Accept Accept Reject (Usury Cap)	`MarketplaceViewModelTest.kt` `testInterestRateBoundaries()`
Collateral LTV (%)	<= 75.00%	Norm: 65.0% Max: 75.0% Max+1: 75.1%	Accept (Normal) Accept (RBI limit) Reject (Over-leveraged)	`CollateralVaultTest.kt` `testLtvCeilingEnforcement()`
Delinquency Grace Period	3 Calendar Days	Day 1: Grace Day 3: Grace End Day 4: Penalty	No penalty fee No penalty fee Trigger <<extend>> UC-15	`RepaymentScheduleTest.kt` `testGracePeriodPenalty()`

6. Requirements Traceability Matrix (RTM)

Direct mapping from IEEE Std 830 Functional Requirements to Implementation & Verification assets:

Req ID	Functional Requirement	Use Case	Code Implementation Artifact	Verification Test Case
FR-AUTH	StrongBox Biometric Authentication	UC-01	`BiometricAuthManager.kt`	`AuthViewModelTest.kt`
FR-KYC	Aadhaar XML & PAN OCR Verification	UC-02	`KycVerificationEngine.kt`	`KycEngineTest.kt`
FR-LOAN	P2P Loan Creation & Auction Bidding	UC-05, UC-08	`MarketplaceRepository.kt`	`MarketplaceViewModelTest.kt`
FR-LEGAL	Section 4 NI Act Promissory PDF & e-Stamp	UC-11, UC-12	`LoanzoPdfEngine.kt`	`LoanzoPdfEngineTest.kt`
FR-REPAY	UPI Autopay & Delinquency Servicing	UC-14, UC-15	`RepaymentRepository.kt`	`RepaymentScheduleTest.kt`
FR-RECOVERY	Order 37 CPC Summary Legal Recovery	UC-18, UC-19	`DisputeRecoveryEngine.kt`	`DisputeRecoveryTest.kt`